Big data analytics workflows with Databricks

Lecture 1 — Introduction to big data

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Big data source

- **human**
  - ~5 new Facebook profiles are created every second
  - ~41.6 M messages are sent on WhatsApp every minute
  - Watching on Netflix for an hour consumes ~ 644 MB of data
- **machine**
  - fitness trackers, cameras, logs, …
- **organizations**
  - transactional data, etc.

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Variety - The different types and sources of data e.g. structured data (databases), semi-structured data (XML, JSON), and unstructured data (emails, documents, videos).

Types of big data

- **structured** (e.g. excel spreadsheet)
- **unstructured** (doesn't fit into a schema, e.g. determine sales revenue given camera footage of transactions, ~90% of all data)
- **semi-structured** (some level of organization, e.g. html code)

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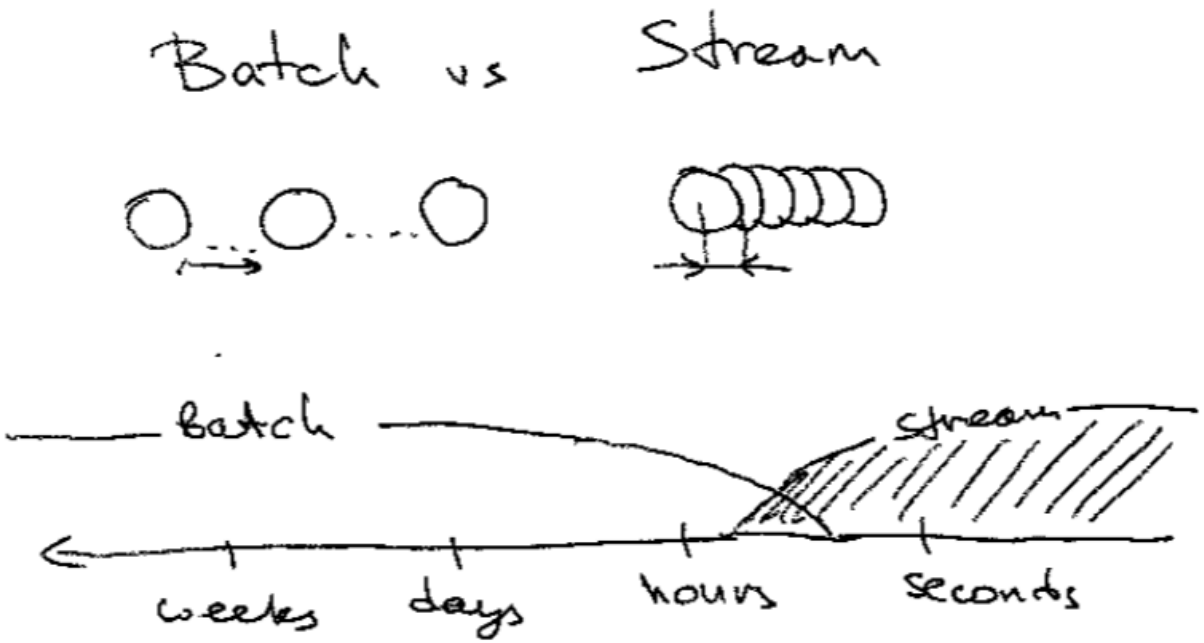
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  e.g. combining structured financial data with unstructured customer reviews
- The ability to present complex data in a format that is understandable and actionable. (Visualization)

Data processing types



- **batch** - e.g. periodic reporting
- **stream** - e.g. fraud detection

Reality …



Implications for Big Data Technologies

Scalable Storage Solutions that can handle large volumes of data efficiently (e.g., distributed file systems, cloud storage).

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Data Integration Methods for integrating diverse data sources and formats (e.g., ETL processes, data lakes).

Common Big Data Storage options

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- **NoSQL Databases** (Apache Cassandra, Mongo, Redis)
- **Data Warehouses** (Redshift, Snowflake)
- **Data Lakes** (built on top of the Hadoop Distributed File System, Amazon S3, etc.)

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- **Lakehouse** (modern data platform architecture paradigm that combines the best features of data lakes and data warehouses)

NoSQL

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Benefits

- **Horizontal Scaling** suitable for handling large volumes of data and high traffic loads
- **Flexible Data Models** & Variety of Data Types easily modify the data structure without downtime
- **Low Latency real-time analytics** and high-throughput applications
- Resilience and **Fault Tolerance** built-in replication features that ensure data redundancy and high availability

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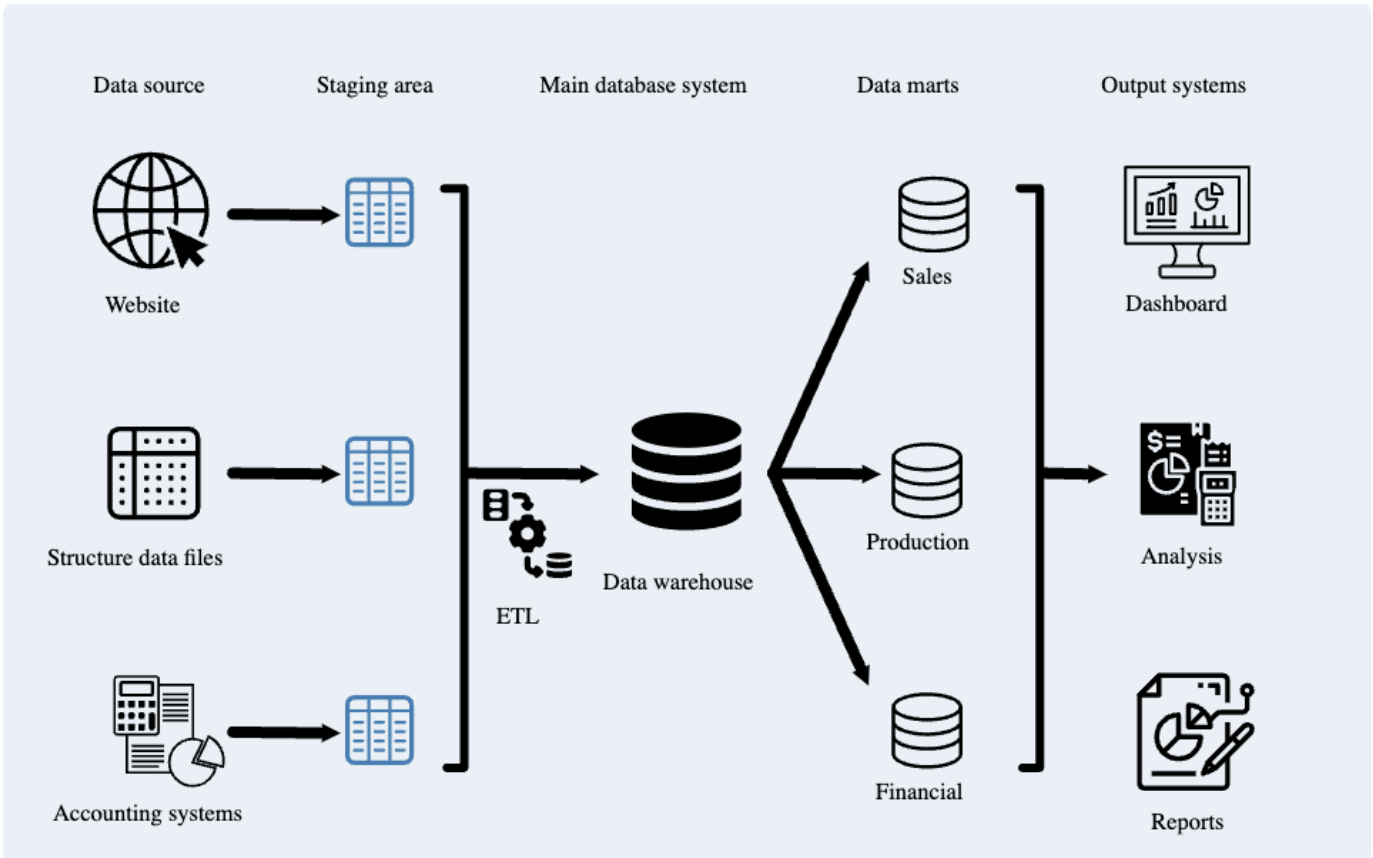
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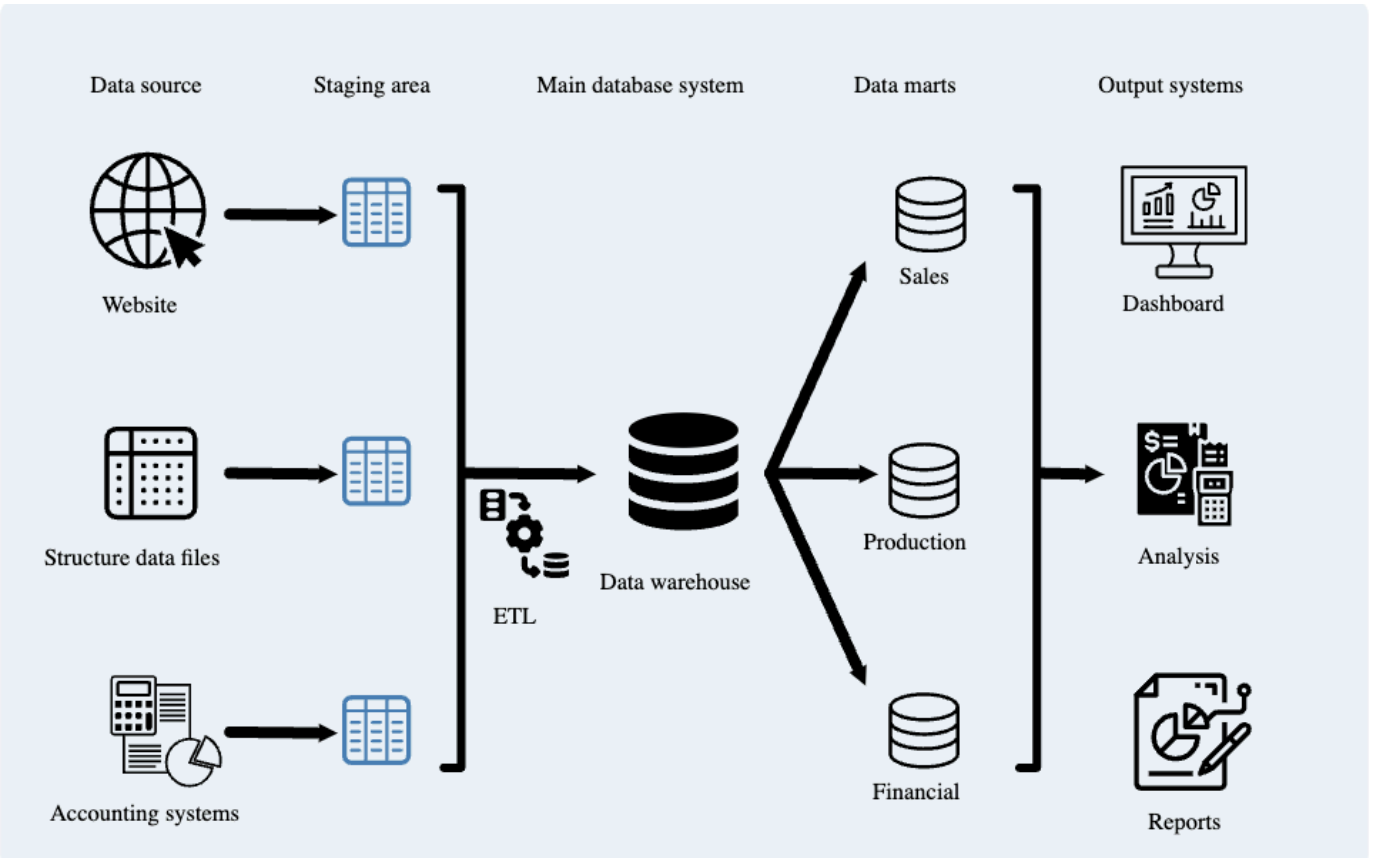
Challenges

- **Limited Transaction Support** do not support ACID (Atomicity, Consistency, Isolation, Durability)
- **Complex Querying** Effective querying may require custom indexing strategies, which can be complex to implement
- **Vendor-Specific Features** Different NoSQL databases have different features and query languages
- **Learning Curve** new data models and query languages

Data Warehouse



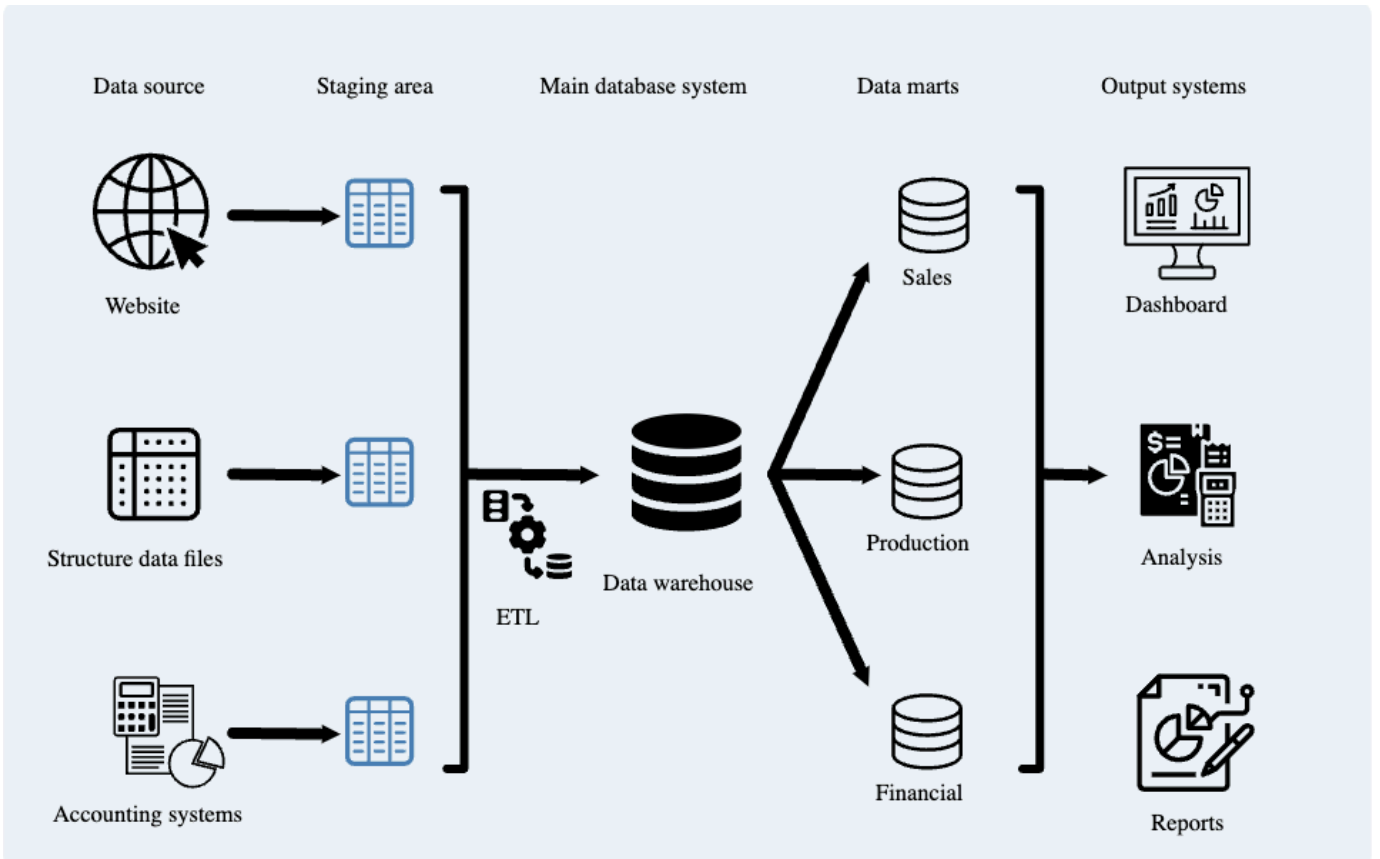
Data Warehouse



Benefits

- reliable
- works well with structured data
- easy to query
- long history of development
- a lot of practitioners
- great for BI applications
- ACID transaction guarantees

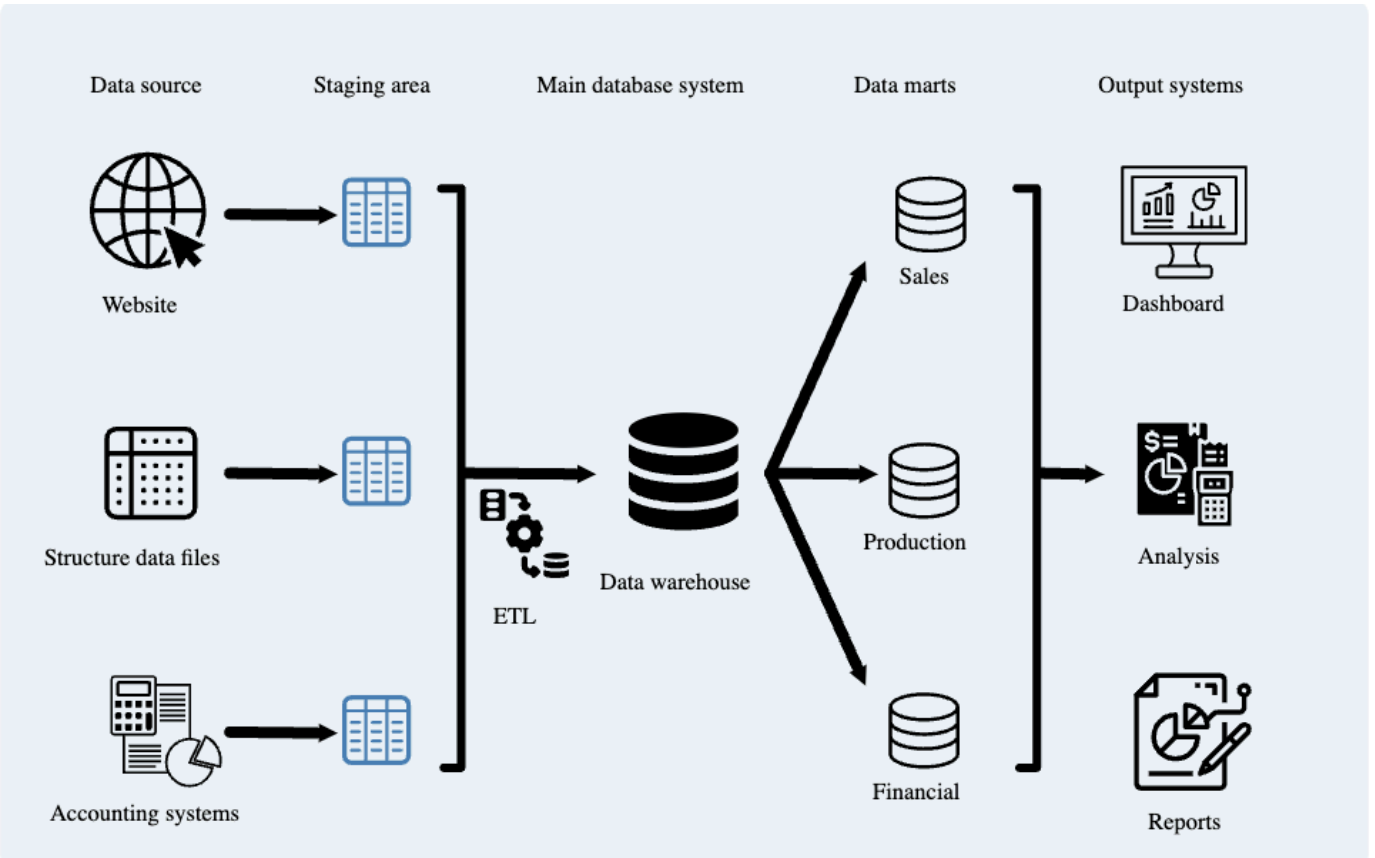
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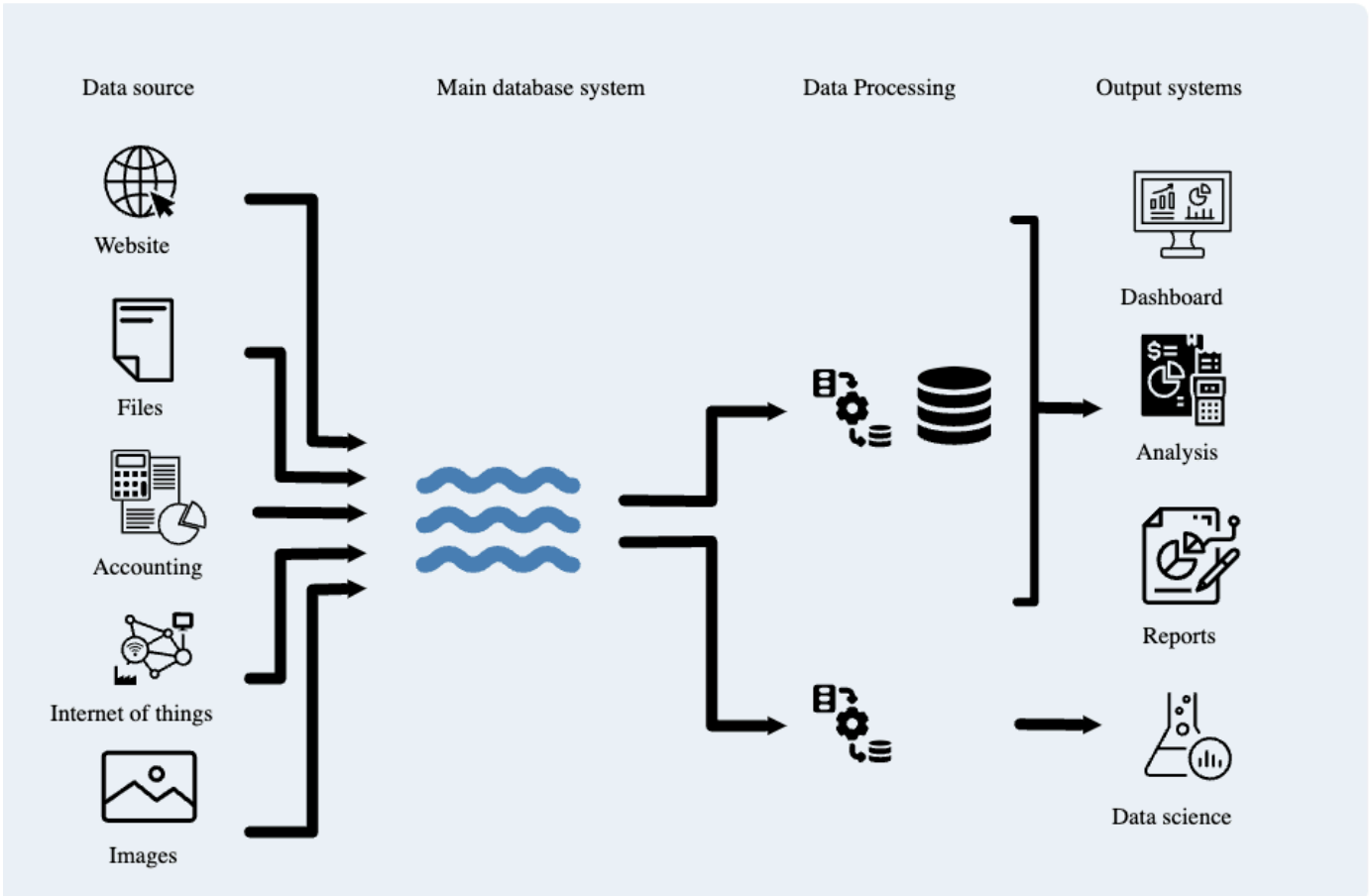
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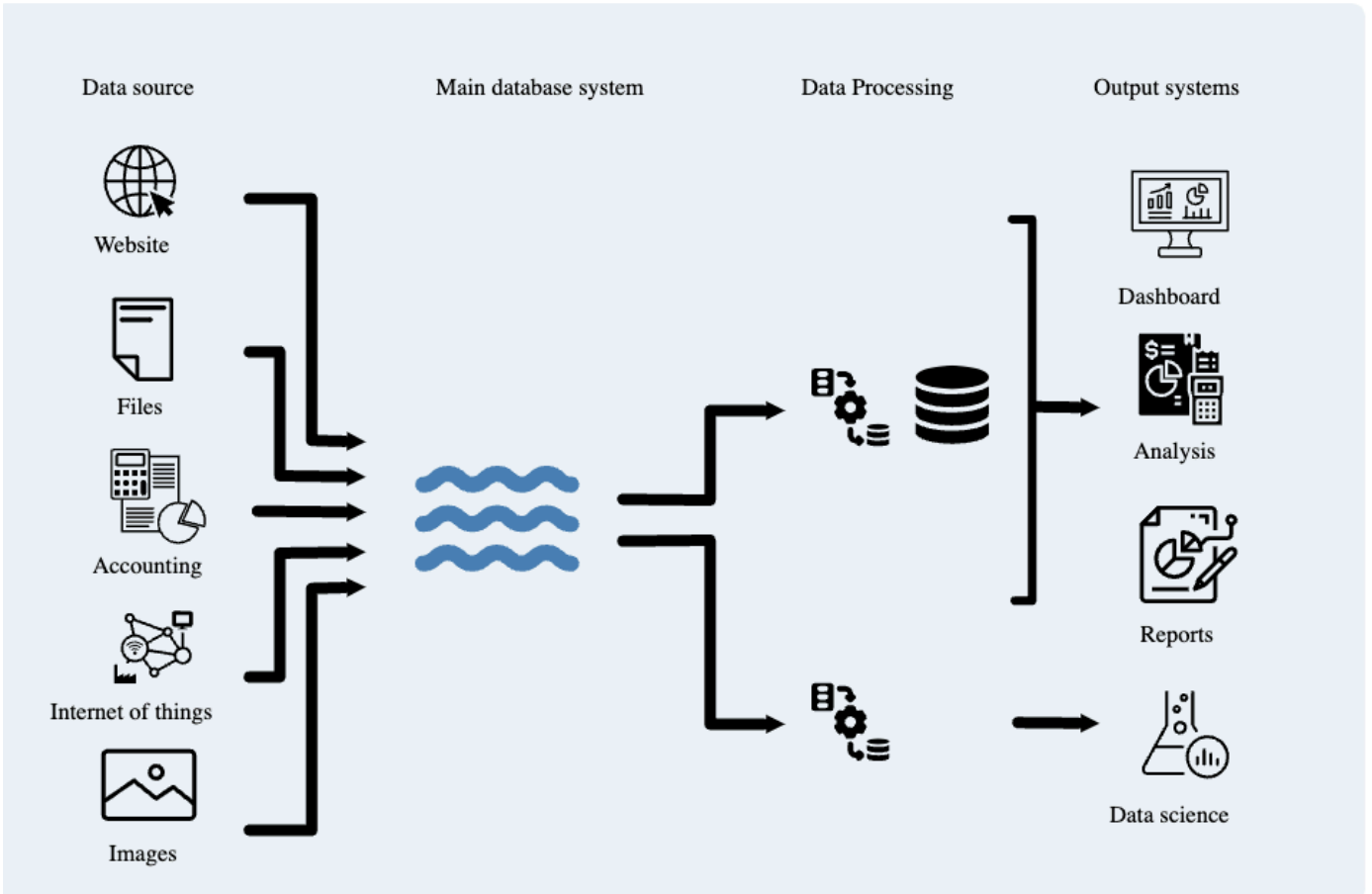
Challenges

- unstructured data
- streaming data sources
- vendor lock-in
- expensive to build and maintain for large volumes

Data Lake


Data source
Main database system
Data Processing
Output systems
Website
Files
Accounting
Internet of things
Images
Dashboard
Analysis
Reports
Data science

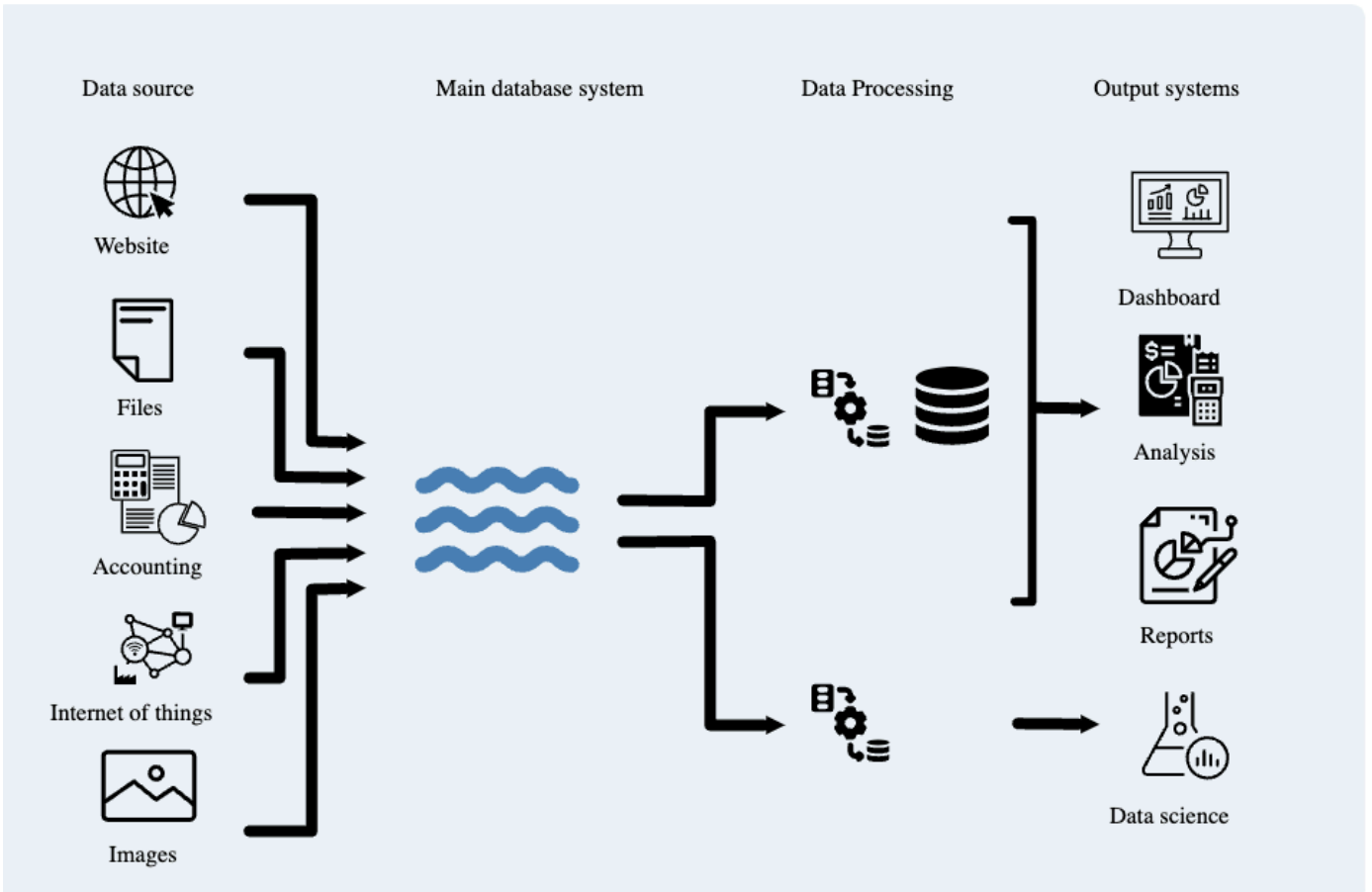
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Benefits

- holds different type of data
- easy to scale
- cheap cloud storage
- separate data storage and compute cost

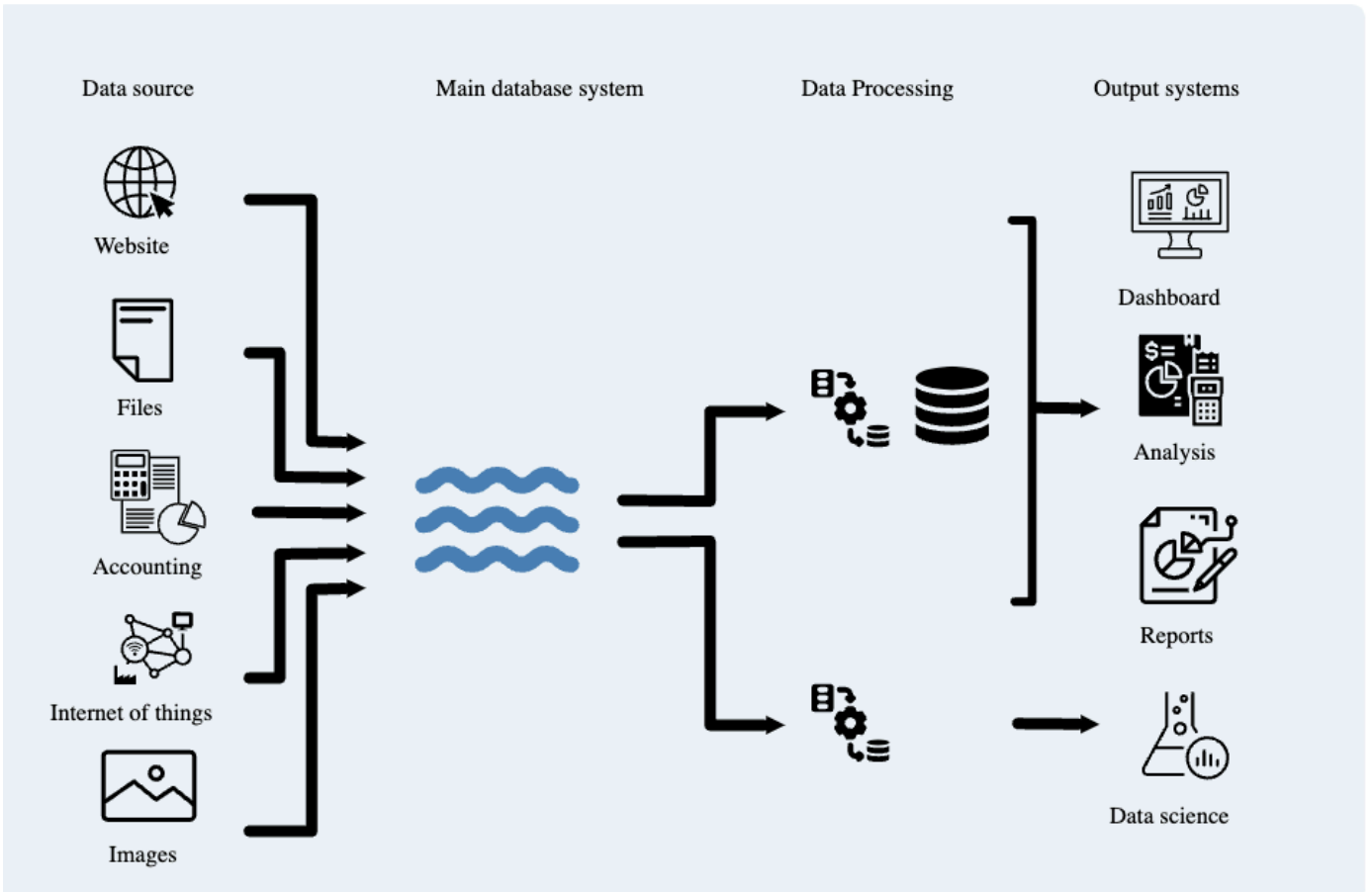
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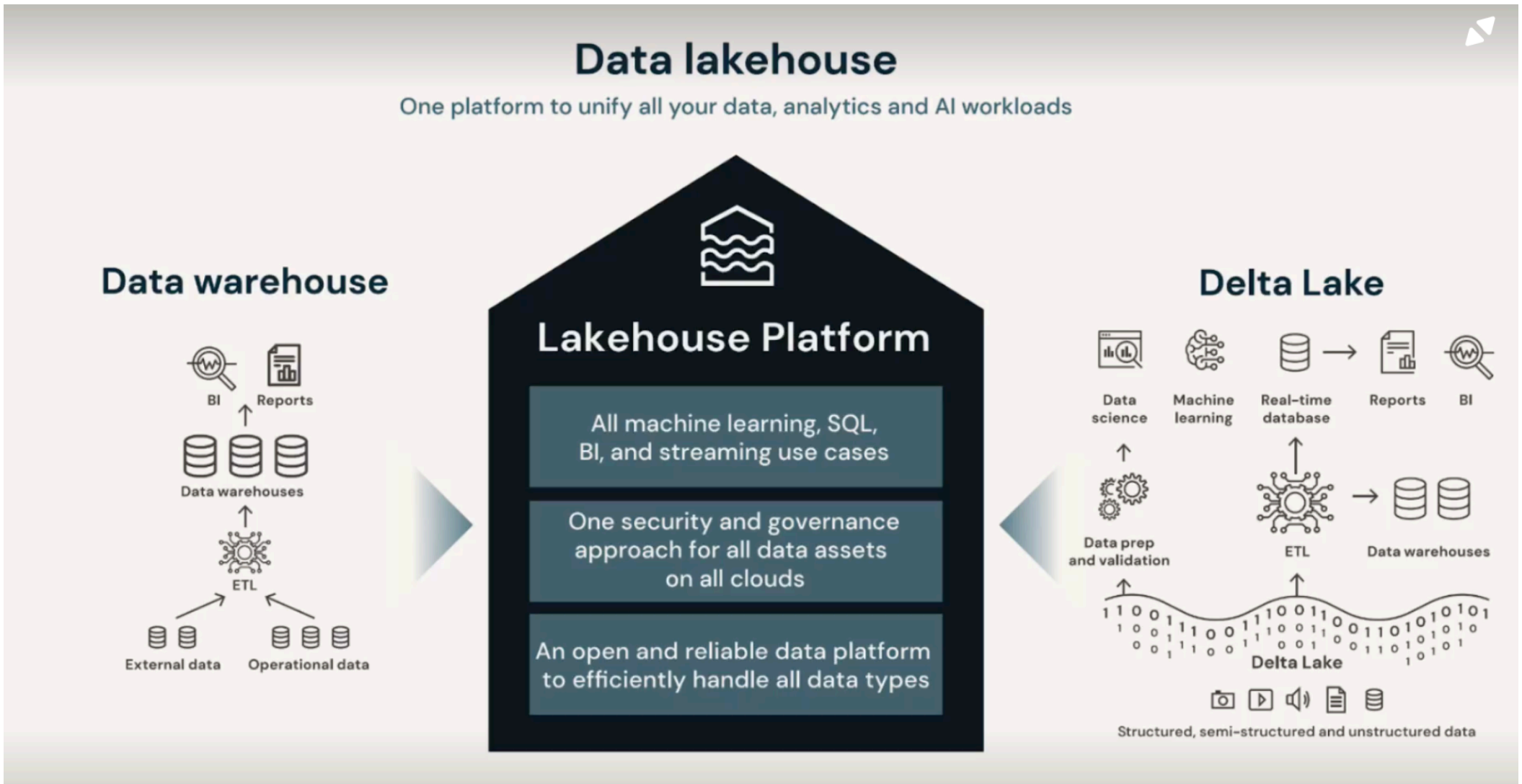
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Challenges

- difficult to understand
- slow query speed
- learning curve
- lack of structure
- do not support transactions
- do not enforce data quality

Lakehouse concept

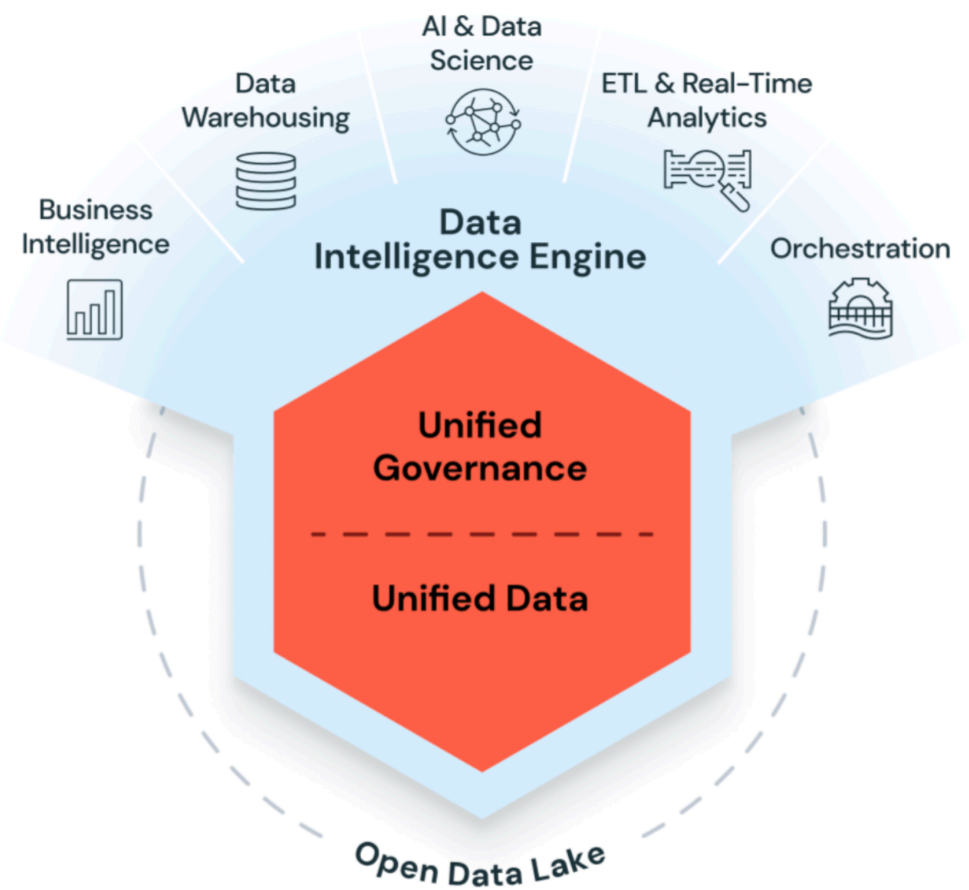


Databricks



What is Databricks?

Databricks is a unified, open analytics platform for building, deploying, sharing, and maintaining enterprise-grade data, analytics, and AI solutions at scale.



Learn more: Databricks intro

Why Databricks?

Figure 1: Magic Quadrant for Data Science and Machine Learning Platforms



Gartner

Learn more: Leader in the 2025 Gartner® Magic Quadrant™ for Data Science and Machine Learning Platforms

Try Databricks

Learn more: Databricks Free Edition

Thank you :)